



NATIONAL INSTITUTE OF TECHNOLOGY PATNA
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
Mid Semester Examination Jan - June 2024

B. Tech: **CSE VIth semester**
Course Code: **CS64121**
Maximum Time: 2 Hours

Course Name: **Cloud Computing**
Max. Marks: 30

Instructions:

1. Attempt all the questions.
2. Assume any suitable data, if necessary.
3. Answer all parts of the question at the same place.
4. **Marks will be deducted for unnecessary writing. Answer all the Questions in brief and precise.**

S.No	Questions	CO	BL	Marks
1.	Discuss the following types of computing environments. Give a comparative analysis of the underlying architecture used in these computing. a. Distributed Computing b. Cluster Computing c. Grid Computing d. Cloud Computing	CO1	L1, L2, L4	2*4
2.	A process in a distributed system runs on one node and accesses data from another node. After some time, for load balancing purposes, this process relocates to a different node. What transparencies should be provided for this process in a distributed system? Explain all such transparencies.	CO1	L2, L3	8
3.	In the process of VM (Virtual Machine) migration to a new host within the same LAN (Local area network), how will the existing communication's messages over the network be re-routed to the new host? Explain in detail.	CO1 , CO2	L2, L3	6
4.	Why is Virtualization needed in a Cloud Computing Environment? Can we use x86 architecture to trap and emulate virtualization? Explain with reasoning. Why?	CO2	L2, L4	4
5.	Discuss the architecture of full virtualization and Hardware-Assisted QEMU/ KVM. Highlight the major differences between both.	CO2	L1, L4	4

*** END ***